Cherry

Traversability analysis for outdoor/offroad scenarios

Fraunhofer IOSB-AST

Department Embedded intelligent systems

Group Cognitive Autonomous Systems

Team Members:

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Our mission: Task 2

- An outdoor robot created a 3D-map of its environment during an exploration phase.
- The environment contains paved but also overgrown and uneven spaces.
- Develop a tool that analyzes the traversability of the 3D-map and visualize the results.
- Note here that the robot has a certain ground clearance and can overcome smaller obstacles.

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Original Point Cloud

Acquired point cloud viewed with CloudCompare v2.13.1

Procedure

Ground plane

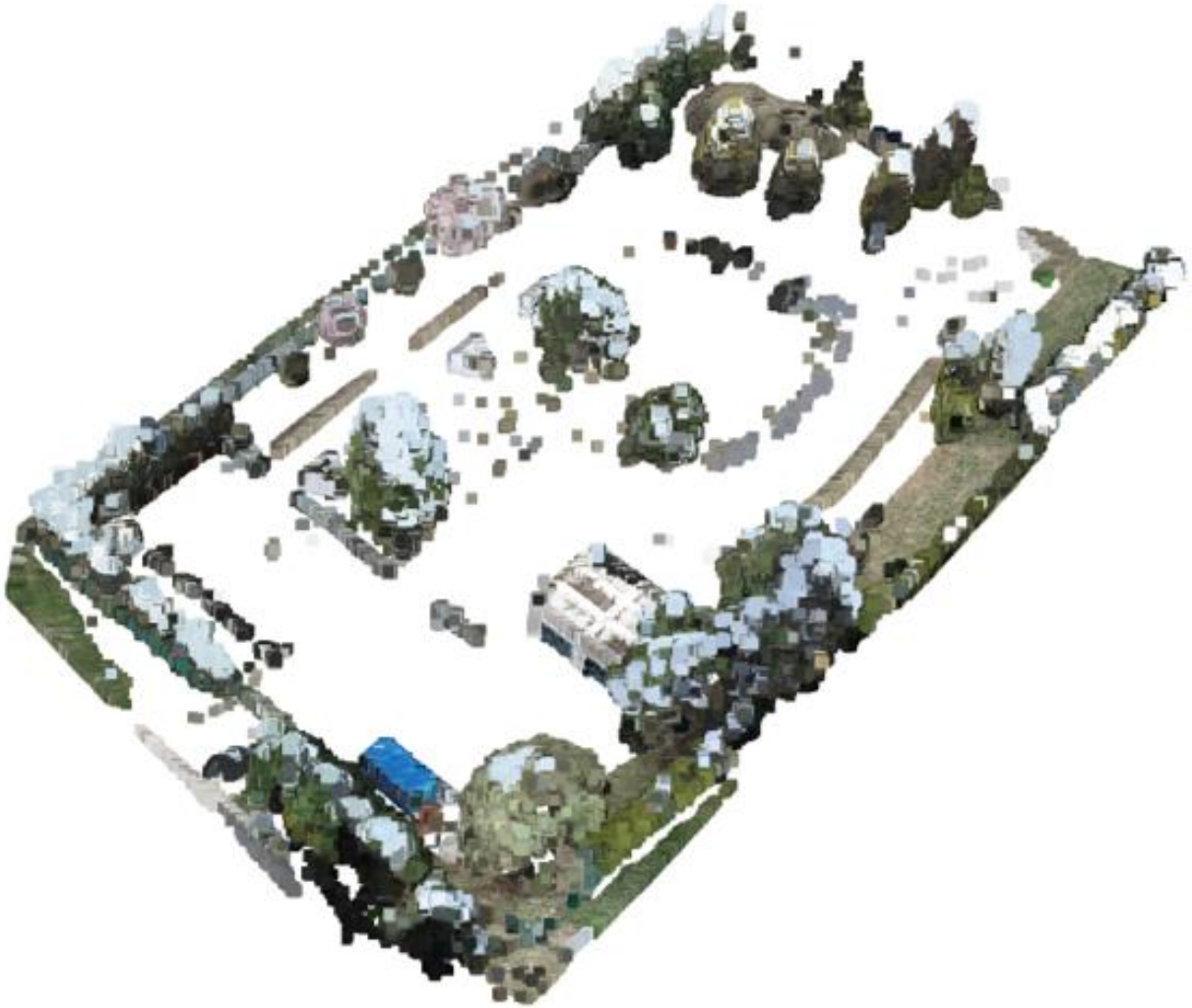
Segmentation of the floor using a plane after a downsampling of the original point cloud:

- $\mathcal{P}$: ax + by + cz + d = 0

Procedure

Possible colliders

The structures that doesn't belong to the ground floor are the possible colliders that the robot have to avoid.



Procedure

Normal Structures

The normal of the structures was obtained, then, the angle of the normal of the structures was compared with the normal of the ground floor.

Procedure

Vertical Objects

Later, the structures that fulfilled the inclination condition with respect to the ground floor were calculated.

Procedure

Robot height

An estimation of the obstacles that the robot may encounter at its height was performed.

Procedure

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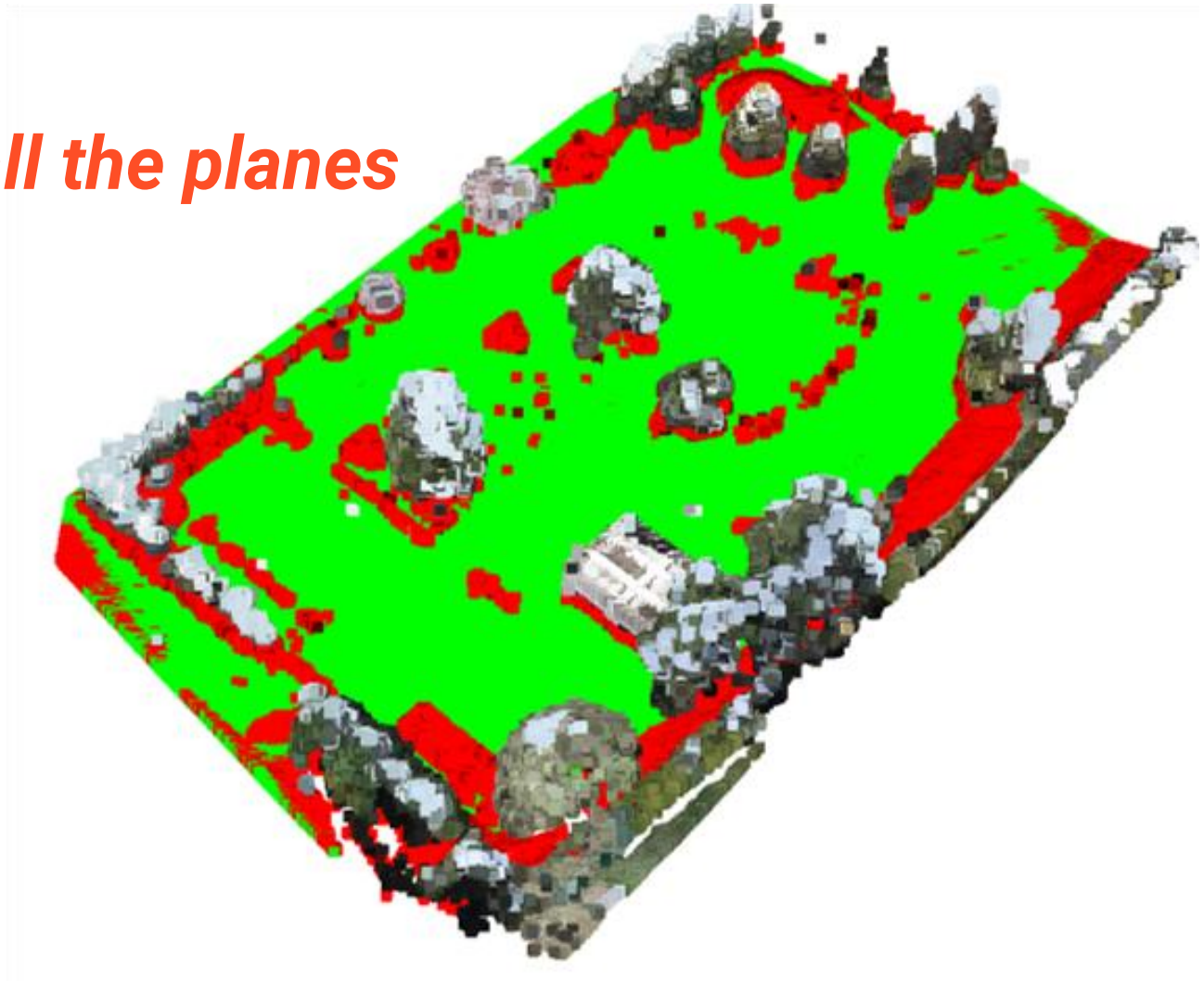
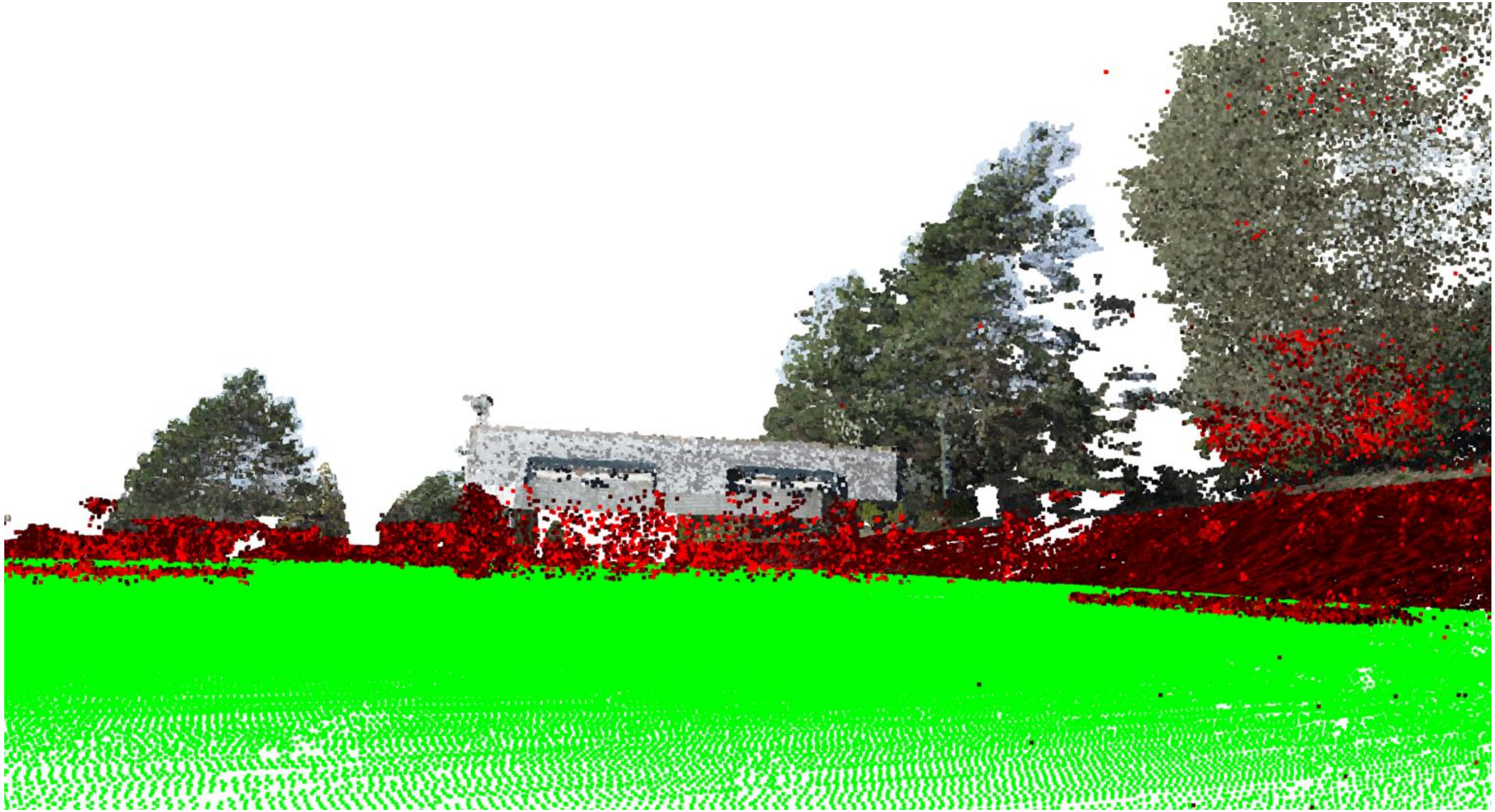
Results

Complete visualization of all the planes

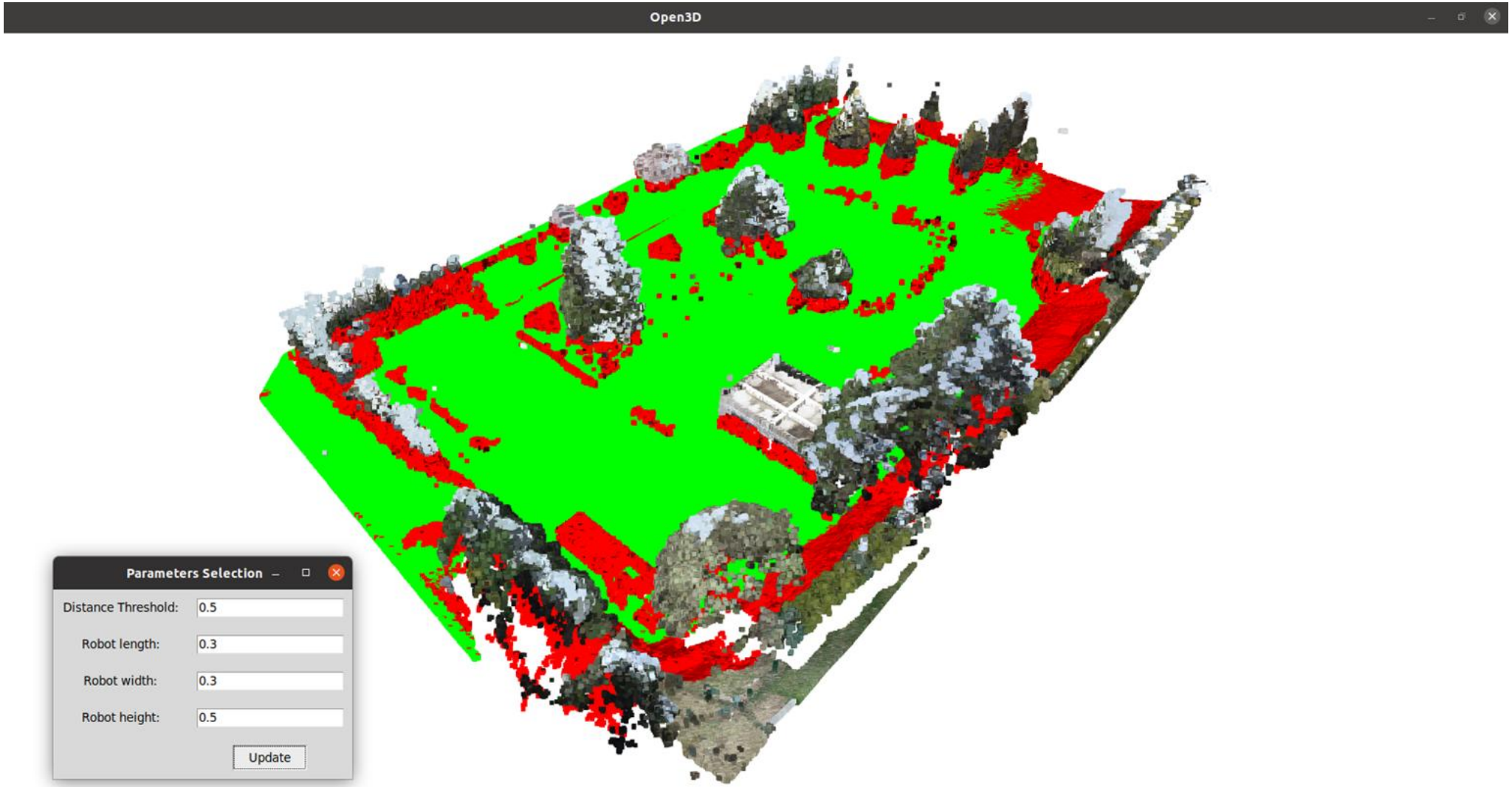
- Ground plane: green
- Obstacles plane: red
- Rest: original point cloud



Results: View of the robot



Simulation


Open3D
Parameters Selection
Distance Threshold:
0.5
Robot length:
0.3
Robot width:
0.3
Robot height:
0.5
Update

Cherry

Thank you for your attention

Your Cherry Team